MUSICAL COMPOSITION WITH NAKED EYE AND BINOCULAR ASTRONOMY

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ABSTRACT

Multimedia compositions that are inspired by outer space usually display and manipulate photographs from telescopes or utilise fractal images or animations. The performer or listener in these cases does not look into space with their own eyes. This paper describes the progress of a musical interface and compositional paradigms that correlate musical performance with naked eye and binocular astronomy, allowing the performer or listener to manipulate the music based on what is being observed in the night sky in real-time.

1. INTRODUCTION

The night sky has been observed by peoples from all continents for millennia. Naked eye astronomy has been used by civilizations to track seasons and to determine the best time to plant and harvest crops. Many of these civilizations; including the Chinese, Greeks, Egyptians and Sumerians; charted roughly familiar patterns of the stars and named them after mythical gods, legendary characters and imaginary creatures. They did this by using a *join-the-dots* methodology, where the pattern was made by creating imaginary lines between the stars or points of light (Massey and Quirk 2012). Other civilizations, such as the Indigenous peoples of Australia, made patterns from the dark areas of the sky (Norris and Norris 2009). Moreover, the stars were used for navigation by ancient mariners, enabling them to accurately traverse the globe (Chittick 2006).

1.1. Visual limitations to viewers

In our modern industrial world, naked eye astronomy has been severely affected by city lights. Many people in urban areas cannot see the Milky Way at night due to the amount of background light from housing, advertising, security and streetlights (Norris and Norris 2009). Only the brightest stars can be seen at night, and the awe of the night sky is now all but lost by many who live in the city (Bee 2006).

Since the advances in astro-photography and the availability of images from the Hubble Telescope in text books, magazines and the internet, many people tend to view stars vicariously. I believe that our technological advancements have in many ways destroyed the personal relationship between the individual and the night sky. Although there are great software programs available for mobile phones, such as *Google Sky,* that enable users to identify stars and constellations in real-time; this research aims to use the information for an artistic outcome rather than just an informative one.

1.2. Existing musical works

Musical composition inspired by or based on outer space is not a new concept. In many cases, the musical composition is based on the superheterodyne manipulation of signals received from radio telescopes (Harger 2011), music performances with slides of deep space astro-images displayed to the audience (Malin and Wesley-Smith 1989), and fractal generated animations that may or may not have any direct correlation to the work itself (Bright 2012). In these cases, the audience is dictated to by the performers as to what they will see. Ironically, the audience is able to hear the music directly as a live performance, but denied the experience of a real-time visual performance of the night sky.

This paper details a compositional paradigm that attempts to spark the interest and stir the imagination of the audience in the night sky by engaging them in a sound installation or concert, where there is a direct correlation between the music and their view of the night sky. This is accomplished through the use of naked eye and binocular astronomy coupled with algorithmic and improvisational compositional techniques.

2. ARTISTIC PERFORMANCE FACTORS

The factors affecting the artistic performance that are addressed in this paper include: light pollution; determining the viewer's focal point in space at a particular moment in time; and using this information compositionally, either as input to an automaton for sound generation, or as stimulus to live musicians.

2.1. Light pollution

Light pollution is essentially background light that limits or obscures what the observer can see (Norris and Norris 2009). The effect of light pollution is different between binocular and naked eye astronomy, and has

both positive and negative effects. Some negative effects can be overcome, while positive effects can be utilised.

2.1.1. Binocular astronomy

The effects of light pollution can be reduced by magnifying the field of view through the use of a telescope or binoculars. This enables the viewer to see stars that are not visible to the naked eye in those conditions (Wallace, Dawes, and Northfield 2012). Although one might consider using a telescope as the interface for the viewer, finding one's way through the night sky for a novice is far easier using binoculars than a telescope, which is "a bit like looking at the sky through a straw" (Wallace, Dawes, and Northfield 2012, p. 10). Binoculars enable the viewer to appreciate open clusters, seeing each star in the cluster in context with the others. A telescope, on the other hand, may provide too much magnification (Bee 2006). The difference between using binoculars and a telescope in the night sky is akin to the difference between observing an insect on a flower with a magnifying glass and observing it with a microscope. The faintest stars visible with the naked eye under dark skies are at magnitude 6; however, a pair of 10x50 binoculars increases this limit to magnitude 9 or 10 (Bee 2006).[1]

2.1.2. Naked eye astronomy

Aside from all the negative effects of light pollution, one positive aspect is that it is easier for a novice to recognise patterns that make up the constellations. This is because there are fewer faint stars in light polluted skies to confuse the viewer (Wallace, Dawes, and Northfield 2012). The area of sky being referenced can be highlighted to the audience through the use of a green laser pointer, which will unambiguously mark the celestial object or point in space. Mounting a green laser to binoculars has two advantages. The first is that the audience can be directed to a point in sky while they are waiting for their turn on the binoculars. The second is that when the audience leaves the performance, they can look up into the night sky and know basically where they were looking, enhancing their memory of the performance (Smith Brindle 1986).

2.2. Mapping the positions of stars in time

Although objects in the night sky move or change position relative to the local horizon throughout the night, it is actually the earth that rotates upon its axis that gives this appearance. For example, a star that appears immediately overhead in Sydney at midnight will not appear overhead in Perth until it is midnight in Perth. Furthermore, due to the Earth's rotation around the sun, the stars each night appear at the positions they did the night before approximately four minutes earlier. The east-west position of a star on the *celestial globe* is called its *right ascension* (RA) and is based on its azimuth at Greenwich Meantime on the vernal equinox. The RA is the *x* axis on star charts and is measured in hours, minutes and seconds (Massey and Quirk 2012). Similarly, the height that a star appears above the horizon is different depending upon the latitude of the viewer. For example, a star that appears immediately overhead in Sydney would appear in the south for a viewer in Cairns. The north-south position of a star on the celestial globe is called its *declination* (Dec.) and is based on its altitude at Greenwich Meantime on the vernal equinox. The Dec. is the *y* axis on star charts and is measured in degrees, minutes and seconds (Massey and Quirk 2012).

It is therefore possible to identify a particular star or point in the in the sky by determining its RA and Dec. and comparing it against entries in a star database. The RA and Dec. can be calculated given the date, time, azimuth and the angle the star appears above the horizon (Duffett-Smith and Zwart 2011).

Many telescopes have a feature called *go-to* that enables users to select a star from an inbuilt database, causing the telescope to automatically slew to the requested star or point in the sky. This is accomplished through the use of gears and rotary encoders (Seronik 2013). A simpler and cost effective alternative, albeit with a lower resolution, is to read the angle and azimuth of the binoculars using triple axis accelerometers and magnetometers through an Arduino microcontroller (Margolis 2011). The azimuth and altitude data is then passed through a Bluetooth serial port to a computer or smart phone to provide further computation.

The output of the sensor unit is converted from horizon to equatorial coordinates by using the date, time and geographic location stored on the computer or smart phone. The magnetic offset between the magnetic and celestial north poles is accomplished by aligning against a bright reference star (Seronik 2013). This resultant RA and Dec. point can now be used as input to the algorithmic music software.

3. COMPOSING WITH RA AND DEC.

Smith Brindle states: "In music, chaos is infinitely tiresome" (Smith Brindle 1986, p. 146). When one considers that people are often in bed sleeping at night when the night sky is at its best, "tiresome" performances should be avoided by composers and performers. Many composers today use algorithmic composition techniques that enable them to map certain physical and conceptual gestures to musical parameters (Winkler 1995). This is possible because excitation and sonification have been separated, allowing for mapping and manipulation, which led to the development of gesture based instruments and responsive environments. This paradigm allows for the input to the instrument to

[1]The magnitude of an astronomical object indicates its brightness. A lower number indicates a brighter object.

come from the output of a series of one or more sensors, which can then be manipulated and mapped to produce sound or music based on those inputs (Fraietta 2006). The RA and Dec. from the sensors are used as inputs to various algorithms that will generate sounds or signals based upon those inputs. There is no *correct* way to compose a work based on this input, nor is there a *correct* style of music. Similarly, there is no algorithm or computational process that could be deemed *correct.* Instead, I suggest different ideas whereby a composition can be based upon creative algorithms that are fed by the sensor input.

3.1. Star map table generation

One method of composition based on a fixed point in space is to create a table of the stars around the celestial point. This table can be created by querying an online database called *VizieR*. Using the RA and Dec. as query parameters, Vizier returns a table of stars with properties such as distance, magnitude, velocity[1] and colour (Ochsenbein, Bauer, and Marcout 2000). For example, if our binoculars were roughly pointed to the star *Acrux,* the RA and Dec. would be approximately 12 hours and -63 degrees. If we queried *VizieR* using these parameters, defining a two degree window, *VizieR* would return a table of over seventy stars with magnitudes[2] ranging from 4.2 to 11. This table can be expressed compositionally or musically in many ways.

Two different works were performed using the *VizieR* star map tables based on the *Crux* constellation. These pieces were experimental compositions to test the viability and practicality of the concept. The first was an improvisation using FM synthesis, while the second was an improvisational stimulus for live musicians. In both cases, the star table map was obtained by inputting the RA and Dec. into the *VizieR* server via a URL such as the following.

```
http://vizier.u-strasbg.fr/viz-bin/asu-
txt?-source=I/311/hip2&-c=186.64975588%20-
63.09905674,rd=2,eq=J2000&-out=RArad&-
out=DErad&-out=pmRA&-out=pmDE&-out=Hpmag&-
out=B-V
```

The parameters to this call are as follows:

`-source=I/311/hip2` indicates the astronomical databases to use. In this case, we are querying Hipparcos catalogue.

`-c=186.64975588%20-63.09905674,rd=2,eq=J2000` indicates RA of 186.64975588[3] in degrees and a Dec. of -63.09905674. This happens to be to location of *ACrux*. The `rd=2` parameter indicates it requests all stars within a two degree radius, while `eq=J2000` indicates that it should use the J2000 coordinate system.

The `-out` parameters define which parameters we want to return, which are: RArad — RA in degrees; DErad—Dec. in Degrees; pmRA and pmDE—proper motion in right ascension and proper motion in declination in milliarcseconds per year; Hpmag—magnitude; and B-V—the colour index.

The output is returned as text that includes a table with each star on a separate line. For example, the first line of tabulated information from that URL is:

```
182.43579591 -62.58184518     -5.90
3.09  8.1840  0.042
```

The returned text is parsed and each line is sent as an Open Sound Control message (Wright and Freed 1997) to the Smart Controller workbench application (Fraietta 2005) for algorithmic processing before sending as MIDI for sound generation.

3.2. Solo synthesiser and binocular realisation

Azimuth and altitude information for the binoculars was accomplished through the accelerometer and magnetometer method, described earlier in the paper, by mounting the circuit to the top of the binoculars in a custom built case, shown in Figure 1. The data was converted to RA and Dec. on the computer and sent as OSC data with a start and stop message to indicate beginning and end of the table data.

The Smart Controller workbench stored all the parameters into a ganged table set that enabled a single index to access the data parameters for a particular star. This effectively made the ganged tables behave as a single table with multiple outputs. The star database parameters were scaled and mapped to MIDI parameters as follows: Dec. to note number, RA to pan, magnitude to velocity, and colour to program change. Proper motion outputs were not used in this work. As each star was output, it would have its own loudness, pitch, stereophonic position, and synthesiser voice. The program changes where linked to five different synthesiser voices, depending on what range the colour value was, thus giving five perceivable colour bands. The stars, being defined by the table indexes, could be cycled using a set range *counter,* which gave a cyclic



Figure 1: Sensor unit mounted to binoculars

[1] Velocity in this case means the proper motions in RA and Dec. per year.

[2] Based on the magnitudes, only ten of the stars in this data set would be visible with the naked eye under dark skies.

[3] Converting hours to degrees is done by multiplying the number of hours by fifteen.

motif. Another method employed was using random indexes to select stars, and random durations between these indexes, which gave an indeterminate effect with wide spatial texture. A subset of star map could be easily selected by placing boundaries to the counter and random number generation. Starting with a lower number of selectable indexes and gradually increasing gave the effect that one experiences when star gazing, where more stars become visible over time (Norris and Norris 2009). It was not difficult to improvise the work using sliders on the software.

Although the work was successful as a study, the work could be realised using a Digital Signal Processing (DSP) sound engine. *Granular synthesis* is a method of DSP sound generation whereby the audio generation is based on small samples of sound, known as *grains*. These grains are multiplied, imitated, and layered, which can be used to create effects such as time stretching and pitch shifting (Xenakis 1992). The composer could choose to map individual grains to each of the stars in the data set, processing them according to properties of the star. For example, stars with higher velocities can be pitch shifted over time according to their speed and direction, or moved spatially through loudspeakers. Similarly, amplitude can be mapped to magnitude, while colour can be mapped to placement in the audio spectrum. The different mapping strategies and possibilities are only limited by the composer's imagination.

3.3. Improvisational stimulus for live musicians

Many composers today recognise that "a player can often create spontaneously something much superior to what can be written down with the crude approximation of notation. But he [or she] must take care that what he [or she] creates builds up into a monolithic whole" (Smith Brindle 1986, p. 146). In such a performance, it is imperative that the musicians are not only able to communicate with one another, but that they are able to work within the intended form of the composition. Although certain styles or genres of music base their "improvisation on a ground bass, on a theme, or using a skeletal harmonic framework such as jazz" (Smith Brindle 1986, p. 11), performers of other styles of aleatoric music are sometimes provided with graphic signs or suggested note groups. Smith Brindle states: "It is all too easy to fall into the trap of wandering on, following the whim of the moment, chopping and changing from sombre meditations to joyous paeans. To the listener, such music is incoherent and boring" (Smith Brindle 1986, p. 12). It is possible to use the algorithmic treatment of the data set derived from computations of the sensor to shape the musical form.

The author collaborated with Colin Bright to perform *sp-A-C--E* for Telescope, Binoculars & Improvising Ensemble (Bright 2013). The performance was musically based on an existing work, *Ground Control to Gliese* (Bright 2012), with the difference being that MIDI triggered samples based on the star maps were also output to the audience and fed to the performers for stimulus for improvisation. A large pair of mounted binoculars and a small telescope were employed to enable the audience the view the sky in real time. Although we originally intended to display images on a video screen from an astro-imaging video camera, this proved too difficult to implement. Furthermore, compared to what people already expect to see from astro-images, the display from a video camera on the screen would have been very unimpressive and was therefore abandoned. Instead, a slide show from images from NASA were cycled through as a slide show and displayed on a screen above the performers.

Although many saw the performance as a success, this was based almost entirely on the skill of the ensemble and their music performance, and virtually nothing at all with the astronomy. The most obvious problem with the performance was that the audience had to leave the room where the musicians were performing. The audience was expected to venture through the back door to go outside in order to view the stars through the binoculars. Consequently, few people ventured outside to the astronomical equipment. Furthermore, the audience could not see the sky and the performers at the same time, so a laser pointer was of no use as a guide during the performance. A second performance was made the following week, however, the sky was completely covered with cloud and it rained, so there was nothing to see through the binoculars. The audience, however, enjoyed the ensemble performance with the NASA image slide show with samples fed from stored star tables.

The success of this type of performance requires a completely new composition, rather than just plugging a pair of binoculars and some MIDI samples into an existing work. The performers and the astronomical equipment need to be under the night sky together for the performance to be experienced as single artistic entity.

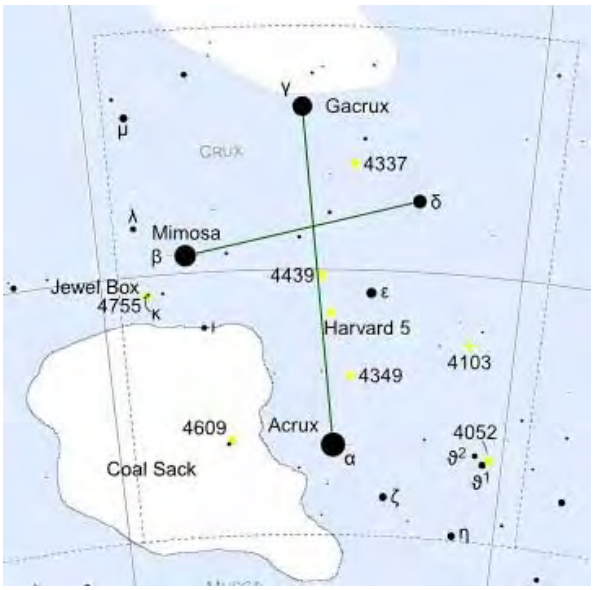


Figure 2: Crux Constellation

Consider a proposed compositional work based on the *Crux* constellation shown in Figure 2, better known as the *Southern Cross* (Massey and Quirk 2012). The focal point is the triple star system *Acrux,* which is used as the initial creative stimulus for the performers. Thematic material is derived from the data set generated from the star map table previously discussed. The data is used to generate MIDI notes, which are fed to the musicians' head phones, similar to how they might use a click track in a recording studio. Alternatively, a live set of motifs can be sent to the performers using a score generating program (Brown 2005). The audience sees the laser is pointing to the southern most star of the Southern Cross, blazing in the midst of the bright *Milky Way,* and when they take their turn viewing through the binoculars, they realise that what they saw as a single star with their naked eye actually appears as two through the binoculars.

The binoculars and laser pointer then move into the large dark nebular that appears to obscure the Milky Way known as the *Coal Sack* (Massey and Quirk 2012, p. 58). "To many Aboriginal groups, the Coalsack [sic] is part of the best known Aboriginal constellation—the Emu in the Sky" (Norris and Norris 2009, p. 5). Instead of providing stimulus from a *VizieR* data set; the performers are fed sounds into their headphones that represent traditional instruments of the indigenous peoples. For example, a drone may represent a didgeridoo, or rhythmic clicks represent clap sticks (Ellis 1985) indicating that they can *free form* with inspiration of the traditional indigenous peoples of that particular area.

Moving out of the *Coal Sack*, the binoculars and laser pointer move into "the pride of the Southern Cross and a favourite of most (southern) amateur astronomers" (Bee 2006 108, p. 108), *NGC 4755,* also "known as the 'Jewel Box' for its varied colours and brightness of the stars" (Massey and Quirk 2012, p. 98). Musically, a complete change in dynamic and style occurs when moving from the *Coal Sack* into to the Jewel Box. Colour spectrum data can be used to generate a MIDI note pitch table, which can be used to create melodic or thematic fragments to both guide and stimulate the musicians.

The work can continue throughout the constellation, moving through to each of the stars and into other *deep-sky objects* such as *NGC 4052;* or possibly the work could include *NGC 4609* in the middle of the *Coal Sack*. The challenge for the composer would be obtaining unity between one section of the work and the next (Smith Brindle 1986), which could be facilitated by an informative narrative that describes the next area of the sky that will be presented top the audience.

3.4. Programmable logic control of musical robots

Some composers today orchestrate for musical robots. These robots generate sound based on the mechanical excitation of a physical object, and are used as alternatives or adjuncts to live musicians or digital synthesis. Control of such robots is accomplished through the use of algorithmic software on personal computers that generate MIDI (Singer et al. 2004). Alternatively, there are also dedicated hardware devices that enable composition and performance based on programmable logic control, the output of which can directly excite musical robots (Fraietta 2006). Although the author has not realised this compositionally in any way using the star data, the concept is quite feasible. For example, star colour from the data set can be used to determine which set of wind chimes or bells are excited. The star count within the data set can also be used to control a loudspeaker loaded with small beads, which, when excited with *control voltages*, create a popping or crackling sound representing a star cluster.

4. CONCLUSION

This paper has described different methods of using algorithmic manipulation to provide compositional stimulus, and meaningful correlation for the audience between what they see and hear. Utilising algorithms that use input from sensors and star charts can provide composers and performers the stimulus needed to create music that has both freedom and form, while simultaneously encouraging audiences to marvel at the wonder of the night sky.

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